$\epsilon_0 = 8.8541878188 \times 10^{-12}$ F/m
$\mu_0 = 1.25663706127 \times 10^{-6} \approx 4\pi \times 10^{-7}$ H/m
$A\cos\left(2\pi f t - \frac{2\pi}{\lambda}x\right) = A\cos(\omega t - \beta x)$
$e^{i\theta} = \cos\theta + j\sin\theta$
$ae^{i\theta} = x + iy$
$(x, y) = a(\cos\theta, \sin\theta)$
$a = \sqrt{x^2+y^2},\ \theta = \arctan\left(\frac{y}{x}\right)$

$\sqrt{\frac{\mu_0}{\epsilon_0}} \approx 120\pi\ \Omega$

$\operatorname{arcosh}(a) = \ln(a + \sqrt{a^2-1})$
$a^2 \gg 1 : \operatorname{arcosh}(a) \approx \ln(2a)$

$\mathrm{Np} = 20\log e\ \mathrm{dB}$
$\mathrm{m/s} = 1.943844\ \mathrm{knots}$

$V = IZ$

Parameter	Coaxial	2-Wire	Parallel-Plate	Unit
R' Resistance	$\frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L' Inductance	$\frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right)$	$\frac{\mu}{\pi}\operatorname{arcosh}\left(\frac{D}{d}\right)$	$\frac{\mu h}{w}$	$\mathrm{H/m}$
G' Conductance	$\frac{2\pi\sigma}{\ln\left(\frac{b}{a}\right)}$	$\frac{\pi\sigma}{\operatorname{arcosh}\left(\frac{D}{d}\right)}$	$\frac{\sigma w}{h}$	$\mathrm{S/m}$
C' Capacitance	$\frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)}$	$\frac{\pi\epsilon}{\operatorname{arcosh}\left(\frac{D}{d}\right)}$	$\frac{\epsilon w}{h}$	$\mathrm{F/m}$

	Propagation Constant $\gamma = \alpha + i\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R'+i\omega L')(G'+i\omega C')}$	$\frac{\omega}{\beta}$	$\sqrt{\frac{R'+i\omega L'}{G'+i\omega C'}}$
Lossless $R' = G' = 0$	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\sqrt{\frac{L'}{C'}}$
Lossless Coaxial	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\left(\frac{60}{\sqrt{\epsilon_r}}\right)\ln\left(\frac{b}{a}\right)$
Lossless 2-wire	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\left(\frac{120}{\sqrt{\epsilon_r}}\right)\operatorname{arcosh}\left(\frac{D}{d}\right)$
Lossless parallel-plate	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\frac{120\pi h}{\sqrt{\epsilon_r}w}$

Field	Meaning	
$\vec{D}$	Displacement Electric Flux Density	$\mathrm{C/m^2}$
$\vec{E}$	Electric	$\mathrm{V/m}$
$\vec{B}$	Magnetic	T
$\vec{H}$	Mag. Intensity	$\mathrm{A/m}$
$\vec{J}$	Current Density	$\mathrm{A/m^2}$

Variable	Letter	Unit	Definition
Frequency	f	Hz	$\frac{u_p}{\lambda} = \frac{f}{2\pi} = \frac{1}{T}$
Wavelength	λ	m	$\frac{u_p}{f} = \frac{\lambda_0}{\sqrt{\epsilon_r}}$
Wavelength in free space	λ_0	m	$\frac{c}{f}$
Angular Frequency	ω	rad/s	$\beta u_p = 2\pi f$
Attenuation Constant	α	Np/m	$\omega\sqrt{\frac{\mu\epsilon}{2}(\sqrt{1+L^2}-1)} \approx \frac{\sigma\eta}{2}$ low loss
Phase Constant	β	rad/m	$\frac{2\pi}{\lambda} = \frac{\omega}{u_p} = \omega\sqrt{\frac{\mu\epsilon}{2}(\sqrt{1+L^2}+1)}$
Permeability Support of mag. fields	μ	H/m	$\mu_r\mu_0$
Permittivity Support of elec. fields	ϵ	F/m	$\epsilon_r\epsilon_0$
Phase Velocity	u_p	m/s	$\frac{1}{\sqrt{\mu\epsilon}} = f\lambda$
Speed of light	c	m/s	$\frac{1}{\sqrt{\mu_0\epsilon_0}}$
Characteristic Impedance Impedance of infinite wire	Z_0, Z_c	Ω	$\sqrt{\frac{L'}{C'}}$
Load Impedance	Z_L	Ω	
Input Impedance	Z_{in}	Ω	$Z_0\left(\frac{z_L+i\tan(\beta l)}{1+iz_L\tan(\beta l)}\right) = Z_0\left(\frac{Z_L+iZ_0\tan(\beta l)}{Z_0+iZ_L\tan(\beta l)}\right)$
Generator Voltage	$\tilde{V}_g$	V	
Input Voltage	$\tilde{V}_{in}$	V	$\tilde{V}_g\frac{Z_{in}}{Z_g+Z_{in}} = V_0^+ + V_0^- = V(0)$
Voltage at a point	$V(x)$	V	$V_0^+e^{-\gamma x} + V_0^-e^{\gamma x} = V_0^+(e^{-\gamma x} + \Gamma e^{\gamma(x-2l)})$
Transmission line length	l	m	
Reflection coefficient at load	Γ		$\frac{Z_L-Z_0}{Z_L+Z_0} = \frac{z_L-1}{z_L+1} = \frac{V_0^-}{V_0^+} = -\frac{I_0^-}{I_0^+}$
Normalized Load Impedance	$z_L, \overline{Z_L}$		$\frac{1+\Gamma}{1-\Gamma} = \frac{Z_L}{Z_0}$
Standing Wave Ratio	S		$\frac{1+\|\Gamma\|}{1-\|\Gamma\|}$
Forward and Backward Voltage wave phasor	V_0^+, V_0^-	V	$V_0^+ = \frac{V_{in}}{1+\Gamma e^{-2\gamma l}}$
Intrinsic Impedance	η	Ω	$\sqrt{\frac{\mu}{\epsilon}}$
Loss Tangent	L		$\frac{\sigma}{\omega\epsilon}$
Skin Depth	δ_s	m	$\frac{1}{\alpha}$
DC Resistance	R_{DC}	Ω	$\frac{l}{\sigma A}$
AC Resistance	R_{AC}	Ω	$\approx \frac{l}{\sigma\omega\delta_s}$ when thin skin

Coaxial	2-Wire	Parallel-Plate
a inner radius	d wire diameter	w width of plate
b outer radius	D wire separation	h height of plate

Law	Formula	Usage
Gauss	$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot ds = Q$
Faraday	$\nabla \times \vec{E} = -\frac{\partial\vec{B}}{\partial t}$	$\oint_C \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial\vec{B}}{\partial t} \cdot ds$
Gauss Mag	$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot ds = 0$
Ampere	$\nabla \times \vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$	$\oint_C \vec{H} \cdot d\vec{l} = \int_S \left(\vec{J} + \frac{\partial\vec{D}}{\partial t}\right) \cdot ds$

L Loss	Classification
< 0.01	Low-loss
$(0.01, 100)$	Quasiconductor
> 100	Good conductor

Z_0 is where you're coming from
Z_L is where you end up
$Z_{in} = \eta_2\left(\frac{\eta_3+i\eta_2\tan(\beta l)}{\eta_2+i\eta_3\tan(\beta l)}\right)$
$\Gamma = \frac{Z_{in}-\eta_1}{Z_{in}+\eta_1}$
$T_{power} = 1 - |\Gamma|^2$

$\vec{D} = \epsilon\vec{E}$ in linear material
$C = \frac{Q}{V} = \frac{\epsilon A}{d}$
$\vec{J} = \sigma\vec{E}$
$\vec{I} = A(\vec{J} + \frac{\partial\vec{D}}{\partial t})$
$\vec{B} = \mu\vec{H}$
$F_m = I(L \times B)$
$\Phi = \int_S \vec{B}\, ds$
$H = \frac{1}{\eta}(\hat{v} \times E)$

$E = \frac{1}{2}CV^2$
$\lambda \gg l$ means ignore transmission line effects

$S_{avg} = \frac{|E|^2}{2\eta_0}$